

Rudra Patel

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EDUCATION

Columbia University

M.S in Artificial Intelligence, AI and Advanced Computing

New York, NY
Expected Dec 2027

University of Ottawa

B.S. in Computer Science, GPA: 3.70

Ottawa, ON
Jun 2026

Relevant Coursework: Introduction to Artificial Intelligence, Fundamentals of Data Science, Probability and Statistics for Engineers, Introduction to Linear Algebra, Information Retrieval and the Internet.

WORK EXPERIENCE

Greet-ly

Founder & Developer

Brampton, ON
Jan 2026 - Jul 2026

- Deployed a Dockerized voice AI platform integrating LLMs, Python, Node.js, n8n, Supabase/PostgreSQL, REST APIs, and webhooks across 10+ workflows.
- Developed LLM-powered agents with tool calling and RAG to ground responses in business-specific knowledge and execute scheduling, insurance, availability, and database operations, increasing call resolution rate from 63% to 84%.
- Validated conversational AI workflows for tool-call accuracy, routing failures, response behavior, and appointment conflicts, enabling the system to handle ~80% of inbound calls and reduce missed calls by ~65%.
- Led pilots across 10 businesses, customizing workflows and converting 2 customers to \$500/month contracts (\$1K MRR) while saving staff ~2 hours/day.

Public Health Agency of Canada (PHAC)

Software Developer (CO-OP)

Ottawa, ON
May 2025 - Aug 2025

- Constructed Python- and SQL-based automation scripts replacing manual data processing workflows, decreasing recurring processing time by 30-40% and decreasing operational error rates by 20%.
- Migrated a manual Excel-based process to Python/SQL ETL pipelines, streamlining ingestion, cleaning, and fuzzy record matching across 10-15 files (thousands of rows) and shrinking daily turnaround ~80%.

Innovation, Science and Economic Development Canada (ISED)

Software Engineer Intern

Ottawa, ON
Sep 2024 - Dec 2024

- Built secure cloud applications using AWS Amplify, Lambda, S3, and SNS, improving response efficiency by 35%.
- Deployed containerized services with Docker and Kubernetes, increasing resource utilization by 45%.

Trillium Health Care Products

Software Development Intern

Brockville, ON
Jan 2024 - Apr 2024

- Automated batch tracking using Azure Functions and Event Hubs, cutting down QA delays by 30%.
- Built demand-forecasting models in Azure ML and automated QA reporting via Azure Cognitive Services, improving planning accuracy by 25% and reducing inspection reporting time by 40%.

RESEARCH EXPERIENCE

Honours Thesis: Diffusion-Based Data Augmentation for HILP

Student Researcher

Ottawa, ON
Sep 2025 - Dec 2025

- Developed and trained a PyTorch denoising diffusion probabilistic model (DDPM) on 3,528 NOAA tornado events, modeling nine spatiotemporal features to augment rare, high-impact events.
- Generated 1,000 synthetic tornado events (+28.3% dataset growth) and evaluated distributional fidelity using KS tests (0.1832), Wasserstein distance (0.0548), and correlation analysis.
- Enhanced downstream rare-event classification F1 score by 10% while preserving physical feature relationships, achieving 0.86 real vs. 0.87 synthetic correlation.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Typescript, Go, C++, C#, Java, HTML/CSS, Node.js, R.

ML/AI: Deep Learning, PyTorch, scikit-learn, NumPy, pandas, LLMs, RAG, AI Agents, Model Evaluation, CI/CD.

Data/Backend: PostgreSQL, REST APIs, ETL Pipelines, Data Analysis, Feature Engineering.

Cloud/Infrastructure: AWS, Azure (Azure ML), Docker, Kubernetes, Linux.

Tools: Git, n8n, React, Angular.